



October 17, 2024

Perfuze Ltd.
Anne-Marie Gannon
Director of Regulatory Affairs
Unit 6, Galway Business Park,
Dangan,
Galway, H91 W7CP,
Ireland

Re: K240711

Trade/Device Name: Zipline Access Catheter
Regulation Number: 21 CFR 870.1250
Regulation Name: Percutaneous Catheter
Regulatory Class: Class II
Product Code: QJP
Dated: September 17, 2024
Received: September 17, 2024

Dear Anne-Marie Gannon:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (<https://www.fda.gov/media/99812/download>) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (<https://www.fda.gov/media/99785/download>).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system>.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory->

[assistance/contact-us-division-industry-and-consumer-education-dice](#)) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Naira Muradyan -S

Naira Muradyan, Ph.D.

Assistant Director

DHT5A: Division of Neurosurgical,

Neurointerventional, and

Neurodiagnostic Devices

OHT5: Office of Neurological and

Physical Medicine Devices

Office of Product Evaluation and Quality

Center for Devices and Radiological Health

Enclosure

Indications for Use

Submission Number (if known)

K240711

Device Name

Zipline Access Catheter

Indications for Use (Describe)

The Zipline Access Catheter is indicated to facilitate the insertion and guidance of catheters into a selected blood vessel in the neurovascular system.

Type of Use (Select one or both, as applicable)



Prescription Use (Part 21 CFR 801 Subpart D)



Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

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510(k) Summary K240711

Submitter Information

Submitter's Name:	Perfuze Ltd.
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Date Prepared:	11 October 2024

Subject Device

Proprietary Name:	Zipline Access Catheter
Common/Usual Name:	Percutaneous Catheter
Classification Name:	Catheter, Percutaneous, Neurovasculature
Regulatory Class:	II
Regulation:	21 CFR 870.1250
Product Code:	QJP

Predicate Device

Proprietary Name:	Route 92 Medical Delivery Catheter
Common/Usual Name:	Percutaneous Catheter
Classification Name:	Catheter, Percutaneous
Regulatory Class:	II
Regulation:	21 CFR 870.1250
Product Code:	DQY
Manufacturer:	Route 92 Medical Inc.
510(k) Number:	K182512

Device Description

The Zipline Access Catheter is a single-use device designed to deliver large-bore catheters to the neurovasculature using standard endovascular techniques. It is supplied sterile and nonpyrogenic. It is available in two sizes: Zipline 088 and Zipline 070. Zipline 088 is designed for use with outer catheters with 0.088" or larger internal diameter, and Zipline 070 is designed for use with outer catheters with 0.070" or larger internal diameter.

During use, the Zipline Access Catheter is advanced over a guidewire to the desired location in the neurovasculature. An outer catheter is then advanced over the Zipline Access Catheter to the desired location.

The Zipline Access Catheter is a single lumen, variable stiffness catheter. It has a tapered flexible tip and a radiopaque marker at its distal end. Proximal to the tapered section is a "bulb" and a radiopaque coil to facilitate bulb visualization under fluoroscopy. The outer diameter of the bulb is designed to closely match the inner diameter of outer catheters, thereby minimizing the gap between the two catheters during navigation.

The Zipline Access Catheter has a hydrophilic coating on the outer surface of the distal end to enhance tracking through the vasculature. A luer fitting located on the catheter hub is used for attachment of ancillary devices. A continuous internal lumen spanning from the distal tip to the proximal hub accommodates guidewires up to 0.018" in diameter.

Indications for Use

The Zipline Access Catheter is indicated to facilitate the insertion and guidance of catheters into a selected blood vessel in the neurovascular system.

Comparison to the Predicate Device

Attribute	Predicate Device Route 92 Medical Delivery Catheter (K182512)	Subject Device Zipline Access Catheter (K240711)
Regulation Number	21 CFR 870.1250	Same
Regulation Name	Percutaneous catheter	Same
Classification	Class II	Same
Product Code	DQY	QJP
Indications for Use	The Route 92 Medical Delivery Catheter is indicated for use with compatible catheters in facilitating the insertion and guidance of catheters into a selected blood vessel in the neurovascular system.	The Zipline Access Catheter is indicated to facilitate the insertion and guidance of catheters into a selected blood vessel in the neurovascular system.
Device Description	The Delivery Catheter is a sterile, single-use, single-lumen, hydrophilically coated, variable stiffness catheter with a flexible, tapered tip.	Same
Principle of Operation	During use, a neurovascular guidewire may be inserted through the Delivery Catheter. The Delivery Catheter is advanced over the guidewire to the target location under fluoroscopy using standard endovascular techniques.	Same
Materials	Biocompatible polymers and metals commonly used in the manufacture of medical devices. Catheter shaft: PTFE, Stainless steel (proximal), and varying durometers of Pebax (distal). Luer: Polycarbonate Marker bands: Tungsten/Pebax	Biocompatible polymers and metals commonly used in the manufacture of medical devices. Catheter shaft: Varying durometers of Pebax, Polyurethane, and Nylon. Luer: Nylon Radiopaque features: Tungsten/ Polyurethane and Platinum/Iridium
Hydrophilic Coating	Present on polymer section of the Delivery Catheter.	Present on distal section of the Zipline Access Catheter.
Inner Diameter	0.019 in.	0.021 in.
Outer Diameter at Bulb Section	0.062 in.	Zipline 070: 0.061 in Zipline 088: 0.080 in
Proximal Outer Diameter	0.028 in.	Zipline 070: 0.063 in. Zipline 088: 0.063 in
Maximum Outer Diameter	0.062 in.	Zipline 070: 0.064 in Zipline 088: 0.080 in
Working Length / Effective length	143 cm	Zipline 070: 158 cm Zipline 088: 147 cm
Packaged Accessories	No accessories included	Same

Attribute	Predicate Device Route 92 Medical Delivery Catheter (K182512)	Subject Device Zipline Access Catheter (K240711)
Sterilization Method	Ethylene Oxide (EO)	Same
Packaging	Nylon/Tyvek pouch inside of an SBS shelf carton.	Polyester/Polyethylene/Tyvek pouch inside of an SBS shelf carton.
Shelf Life	8 months	12 months

Non-Clinical Testing

Biocompatibility Testing

The Zipline Access Catheter is constructed using materials that are commonly used in the medical device industry. All patient-contacting components have been evaluated for biocompatibility in accordance with ISO 10993-1, "Biological Evaluation of Medical Devices Part 1: Evaluation and Testing within a Risk Management Process". The Zipline Access Catheter has been classified per ISO 10993-1 as an externally communicating device that contacts circulating blood for a limited (< 24 hours) duration. A summary of the biocompatibility testing is outlined below.

Test	Results
Cytotoxicity – ISO Elution Method	The test article is non-cytotoxic.
Sensitization – ISO Guinea Pig Maximization Sensitization Test	The test article did not elicit a sensitization response.
Irritation – ISO Intracutaneous Reactivity	Requirements of the ISO intracutaneous reactivity test were met for the test article.
Acute Systemic Toxicity – ISO Acute Systemic Injection	Requirements of the ISO acute systemic injection test were met for the test article.
Material-Mediated Pyrogenicity	The test article is non-pyrogenic.
Hemocompatibility – Complement Activation (SC5b-9)	The test article is not considered to be a potential activator of the complement system.
Hemocompatibility – Partial Thromboplastin Time	The test article is not considered to be an activator of the intrinsic coagulation pathway.
Hemocompatibility – ASTM Hemolysis	The test article is considered non-hemolytic.
Hemocompatibility – Thromboresistance	The test articles have similar thromboresistance characteristics as the control devices.

Performance Testing - Bench

The successful completion of the performance testing listed in the following table demonstrates that the Zipline Access Catheter is suitable for its intended use.

Test	Test Method	Conclusions
Visual Inspection	Device surface characteristics were assessed to confirm freedom from defects that could cause injury.	The device surface characteristics are suitable for its intended use.
Dimensional Inspection	Device dimensions were measured to confirm conformance to the specifications.	The device met established specifications.
Buckling Force Test	Test specimens were tested for buckling force at the distal tip.	The device met established specifications.
Simulated Use Testing	Deliverability and compatibility with ancillary devices were evaluated in a neurovascular model.	The device performs as intended under simulated use conditions.
Torque Strength	The test specimens were rotated with the distal end constrained from movement to evaluate integrity after rotation.	The device met established specifications.

Test	Test Method	Conclusions
Torque to Failure	Catheters were rotated at the proximal hub with the distal end constrained until device failure.	Resistance to torque failure was similar to control devices.
Hydrophilic Coating Integrity	The integrity of the hydrophilic coating was evaluated after multiple insertion and withdrawal cycles.	The hydrophilic coating integrity is suitable for its intended use.
Tensile Strength	The tensile strength was evaluated for the bonds between sections of the catheter and at the distal tip.	The device met established specifications.
Air Leakage	Tested per ISO 10555-1:2013 Annex D.	The device integrity is suitable for its intended use.
Liquid Leakage	Tested per ISO 10555-1:2013 Annex C.	The device integrity is suitable for its intended use.
Luer Dimensions and Performance	The luer was evaluated per ISO 80369-7:2021 and ISO 80369-20:2015.	The device met established specifications.
Kink Resistance	Test specimen segments were formed into a defined bend diameter to evaluate kink resistance.	The device met established specifications.
Radiopacity	Radiopacity of the device was evaluated in an animal model under fluoroscopy.	The radiopacity was similar to a control device.
Particulate Recovery	The purpose of this test was to quantify the particulate sizes and counts generated during simulated use of the test article.	The particulate generation was similar to control devices.

Animal Testing

The *in vivo* performance and safety of the subject device was assessed in suitably sized branches of the carotid arteries in a porcine model in comparison to a control device (Scout Introducer, provided with the AXS Vecta Intermediate Catheter). The study was conducted under Good Laboratory Practices. Assessments included the maximum number of delivery and retrieval attempts expected during a clinical procedure with the Zipline Access Catheter. Angiographic assessments during the procedure and prior to necropsy demonstrated acceptable results. Animals were survived to a subacute (3-day) time point to evaluate the vascular response to device use. Histological evaluations concluded that artery segments in which the subject and control devices were used exhibited a comparable tissue response.

Sterilization

The Zipline Access Catheter is sterilized using a validated EO process with a sterility assurance level of 1×10^{-6} . The validation was conducted using the overkill method according to ISO-11135, "Sterilization of Health-Care Products - Ethylene Oxide - Requirements for the Development, Validation And Routine Control Of A Sterilization Process For Medical Devices".

Shelf Life and Packaging

The Zipline Access Catheter has a 12-month shelf life. Accelerated aging testing based on ASTM F1980 was conducted to verify packaged device performance. Device performance was also verified by functional and performance testing on devices conditioned by accelerated aging.

Clinical Data

The non-clinical performance testing was determined to be sufficient to support the substantial equivalence of the subject device to the predicate. Therefore, no clinical study was conducted.

Conclusion

The intended use of the Zipline Access Catheter is the same as that of the predicate device. The Zipline Access Catheter and the predicate device use the same operating principles and have a similar design. The technological differences identified do not raise different questions of safety or effectiveness. The successful completion of biocompatibility, bench performance, and animal testing demonstrates that the Zipline Access Catheter is substantially equivalent to the predicate device.